RAKESH NARAYAN IYER

Staff Software Engineer
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EXPERIENCE

Anchorage Digital - Member of Technical Staff

Blockchain Protocols | Mar 2025 - Oct 2025

- **High Throughput Scanning Infrastructure:** Led the initiative to build a streaming platform for blockchain transaction and ledger scanning, targeting high throughput and faster recovery.
- **Unified Ledger Data:** Added support for Unified Ledger Data for Solana, Bitcoin, and Filecoin; now driving the next phase, producing accurate ledger data for supported chains using reliable distributed stateful protocols.
- **Operational Excellence:** Led high-throughput Fogo blockchain bring-up and drive incident resolution, performance improvements, and complex bug fixes across blockchain services.
- **Technical Leadership:** Provide support for the Unified Ledger Data initiative; guide engineers on scalability, reliability, and architecture, while onboarding, coaching, and mentoring engineers across the organization.

Google - Staff Software Engineer

Distributed Threat Detection | Aug 2022 - Nov 2024

- **Threat Detection Engineering:** Implemented kernel rootkit detectors and enrichers, and built Malicious File scanning support ingesting rapidly evolving threat intelligence.
- **Distributed Infrastructure:** Provided distributed systems and infrastructure expertise for the threat detection organization, and executed horizontal initiatives including data residency and universal access lockdown.
- **Engineering Excellence:** Improved system scalability, performance, maintainability, and testability through scaling fixes, performance optimizations, and code refactoring, earning the C++ readability reviewer badge.
- **Technical Leadership:** Reviewed organization-wide design documents, incident postmortems, and operational processes, while presenting talks on systems, technologies, and complex investigations.
- **Research & Innovation:** Architected an early version of the threat detection service in a multi-cloud environment and built a proof-of-concept File Introspection detector.

Twitter - Staff Software Engineer

Real-Time Storage, Twitter Health Data | Jul 2018 - Aug 2022

- **Privacy & Data Protection:** Led privacy and data protection for the company-critical KV storage system, encompassing data erasure, service authentication, data governance, and data retention; led 8 engineers.
- **Data Erasure:** Built a comprehensive, fully managed Data Erasure service from the ground up, supporting all customer use cases.
- **ML Data Infrastructure:** Served as Technical Lead for a 10-person team managing Prevalence, FPR, and other ML metrics pipelines; co-designed False Discovery Rate measurement using Apache Airflow and guided the ingestion infrastructure for a BigQuery-backed data warehouse.
- **Datacenter Migration:** Played a key role in implementing the migration of the KV storage system, data, and customers from ATL to PDX datacenters.
- **Cross-Functional Leadership:** Collaborated with data scientists, technical leads, PMs, TPMs, and EMs to shape product roadmaps.

Amazon - SDE II

DynamoDB, WorkDocs | Mar 2015 - Jul 2018

- **Distributed Systems & Scale:** Enhanced DynamoDB through multiple scaling improvements, enabling the use of faster hardware and increasing throughput by 3x.
- **Reliability & Performance:** Redesigned the distributed replication client for the WorkDocs collaboration service, improving reliability, availability, fault tolerance, and performance.
- **Distributed Algorithms:** Invented a distributed transactional algorithm for efficient and stable document and folder movement.
- **Operational Excellence:** Resolved critical production issues, created operational tooling and dashboards, and enabled load simulation for rapid debugging and fix validation.

Riverbed - Senior Software Engineer

Steelhead Platform | Feb 2014 - Mar 2015

- **Container Infrastructure:** Designed and developed a private container infrastructure service managing compute, disk, and networking to interface with a custom high-speed routing switch.
- **Performance Scaling:** Containerized the Steelhead WAN optimizer, increasing performance by 1.5x.

NVIDIA - Senior Software Engineer

Tegra Platform, Storage Platform | Jan 2006 - Feb 2014

- **Mobile Systems:** Developed display system software for Android-based Tegra phones and tablets, including display control and power management features for improved idle behavior.
- **Linux & Embedded Systems:** Developed open-source Linux kernel and U-Boot keyboard drivers for ChromeOS netbooks, and designed features for mobile portability, power management, and error recovery.
- **Enterprise Storage:** Developed RAID 0/1/5 and Windows storage software for enterprise nForce servers, and designed and developed offline experiences for home and enterprise environments.

TECHNICAL SKILLS

Systems & Infrastructure: NoSQL Databases, ML Infrastructure, Distributed Filesystems, Compute Infrastructure, Kernel & Embedded Systems Programming

Data & Streaming: Apache Beam, Google Dataflow, Apache Hadoop, Apache Airflow, BigQuery

Languages: Go, Java, Python, C++, Scala, C

AI-Assisted Development: Claude Code, Google Gemini

EDUCATION

Stony Brook University - M.S., Computer Science · GPA: 3.85/4.0

Pune Institute of Computer Technology, India - B.E., Computer Engineering · First Class

PATENTS

Updating Records In A Real-Time Storage System · Distributed Health Manager Using Peer Leases · US8516190 · US8055842

RESEARCH PUBLICATIONS

Operating System Profiling via Latency Analysis · On Incremental File System Development · Versatile, Portable, and Efficient OS Profiling via Latency Analysis